

An Explainable AI Framework with ML Model Stacking for House Price Prediction

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Abstract—The post-COVID-19 real estate market has presented challenges in finding affordable housing. Accurate house price prediction systems can aid in addressing this issue. This research develops a stacking regression model framework for improving predictive performance with the consideration of Explainable AI (XAI) of house price estimation. This new model was subsequently compared to Regression-based prediction models and a Neural Network to assess model performance. The Stacking Regressor's strong r^2 value of 0.901 and its noticeably lower MAE and RMSE highlight strong precision. XAI techniques, particularly SHAP values, were employed to enhance model interpretability, offering insights into individual feature contributions.

Keywords—ensemble modeling, stacking models, house price prediction, explainable AI, regression

I. INTRODUCTION

Recent economic disruptions, including the COVID-19 pandemic and the ongoing geopolitical conflict in Ukraine, have exacerbated volatility in the housing market. The Russia-Ukraine conflict has significantly impacted the global economy, contributing to a 2% increase in global inflation for 2022, with anticipated residual effects [1]. Concurrently, the pandemic has led to a contraction in housing supply and heightened price distortions due to government-imposed shutdowns [2]. During these shutdown periods, housing prices surged by approximately 2.5%, particularly affecting lower-priced segments of the market, with broader implications as supply constraints and demand pressures persisted [2]. Post-shutdown pricing dynamics varied by property density and market conditions, further complicating market behavior [3].

The limitations of established real estate platforms, such as Redfin and Zillow, are well-documented. These platforms often face criticism for biases and inaccuracies in their predictions, particularly in volatile and competitive markets [4]. This highlights the necessity for advanced predictive models that can effectively navigate the complexities introduced by recent disruptions and market imbalances.

Existing research underscores the complex interplay between market forces and regulatory responses during economic disruptions. For instance, studies have demonstrated

that housing prices initially rose during government-imposed shutdowns due to reduced supply, with significant variations across market segments. In the Fargo-Moorhead-West Fargo MSA, prices increased by about 2.5% during the shutdown, primarily impacting lower-priced homes. Subsequent price increases were observed more broadly due to both diminished supply and elevated demand [2]. Moreover, research indicates that post-shutdown pricing effects were influenced by property density, with suburban areas experiencing price increases and densely populated areas seeing declines [3].

These market distortions emphasize the need for a way to aid consumers who are interested in purchasing or investing in real estate. Common implementations of housing price estimators need to be more reliable and accurate housing price predictions. The aftermath of the 2008 financial crisis highlighted issues related to inflated market prices through cash-based transactions and development strategies, complicating fair property valuations [5]. Addressing these challenges, this research leverages advanced machine learning techniques, specifically Explainable Artificial Intelligence (XAI) integrated with a multi-level stacking ensemble approach, to improve prediction accuracy and transparency in housing market forecasting [6].

Stacking models, which aggregate outputs from multiple base models, have shown superior machine learning performance in various domains, including disease detection and hospital readmissions [6], [7]. This study introduces an innovative stacking ensemble framework for house price prediction, employing supervised learning techniques such as Random Forest Regressor (RFR), Neural Networks (NN), Support Vector Regression (SVR), and Gradient Boosting Regression (GBR). A key advancement is the integration of XAI to ensure model transparency and interpretability, thereby enhancing stakeholder confidence and providing insights into feature impacts [7], [8]. No previous works considered the use of an explainable stacking model for house price prediction, however simply explainable models or stacked models have been implemented.

By building on advancements in XAI and multi-level stacking models, this research aims to offer a robust solution to the challenges of predicting house prices amidst market

uncertainties, a platform contributing to more equitable and informed housing market practices.

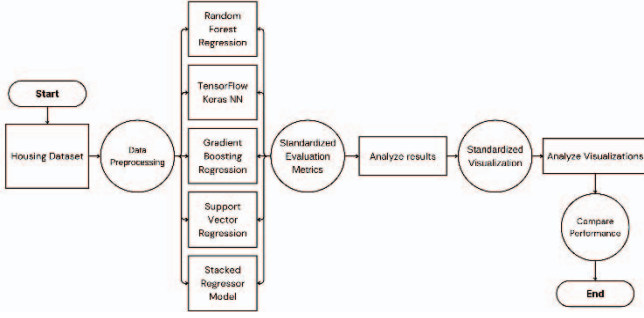


FIG. 1. Workflow: model implementation, visualization, and evaluation.

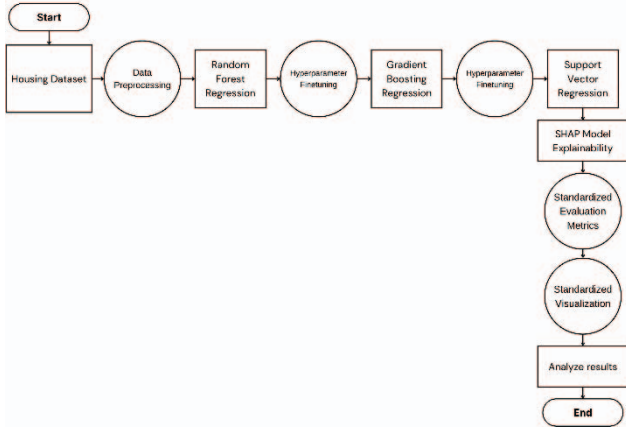


FIG. 2. Explainable Stacked Regressor Model Architecture

II. MODELS AND METHODOLOGY

A. Programming Environment

All computations, model executions, and data analyses were performed using Python, executed within the Google Colaboratory (Colab) IDE. This choice ensures standardized performance, facilitates code reproducibility, and simplifies computational resource management. The following Python libraries were utilized to support the workflow:

- **Pandas:** For data manipulation and preprocessing.
- **NumPy:** For numerical operations and array handling.
- **Matplotlib:** For data visualization and graphical representation.
- **Sci-Kit Learn:** For implementing machine learning algorithms and model evaluation.
- **TensorFlow Keras:** For constructing and training neural network models.
- **SHAP:** For model interpretability and understanding feature impacts in the XAI model

B. Data and Preprocessing

The primary dataset for this study was obtained from Kaggle and comprises residential property sales data from Ames, Iowa, USA, spanning the years 2006 to 2010. This dataset includes

approximately 1,460 data points and 79 features, capturing various aspects of property characteristics. The target variable for this analysis is the sales price of the properties in USD. The data set contains both numerical and categorical variables, facilitating a comprehensive analysis of property price determinants.

To ensure the robustness of the machine learning models employed, the dataset was partitioned into training and testing subsets, following an 80/20 train-test split. This approach aligns with the standard practice of using 80% of the data for training and 20% for testing to evaluate model performance and prevent overfitting.

Data cleaning methods were further implemented. First, we used dummies to fill in missing values in the columns and prepare the dataset for the regression models. This process also included using Simple Imputer to convert the categorical columns into numerical values for the models to understand the text-based information. Additional Feature Engineering was utilized to combine variables that needed to be grouped together. This reduces the model complexity and provides further insight as to how the model predicts values. Additionally, we scaled the data and implemented hyperparameter fine tuning to get the best predictive performance out of each of the models.

C. Prediction Models

Four supervised ML models were implemented on the dataset to serve as a benchmark for comparison of predictive performance. The models employed in this study encompassed Random Forest Regressor (RFR), TensorFlow Keras Neural Networks (NN), Gradient Boosting Regressor (GBR), and Support Vector Regression (SVR) due to their common use in previous works in this field. Each model was trained and evaluated using the same preprocessed data to maintain consistency and evaluated using standardized evaluation and visualization functions (see workflow in Fig. 1).

The new Stacking Regressor was created using three regression models: RFR, GBR, and SVR. These three models were stacked to integrate the outputs of the individual models, aiming to leverage their combined strengths for enhanced prediction accuracy (Fig. 2). GridSearchCV was utilized to fine-tune hyperparameters and optimize model performance.

The performance of the models was assessed based on three key metrics: Root Mean Squared Error (RMSE) (1), Mean Absolute Error (MAE) (2), and r^2 score (3).

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (1)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2)$$

$$r^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

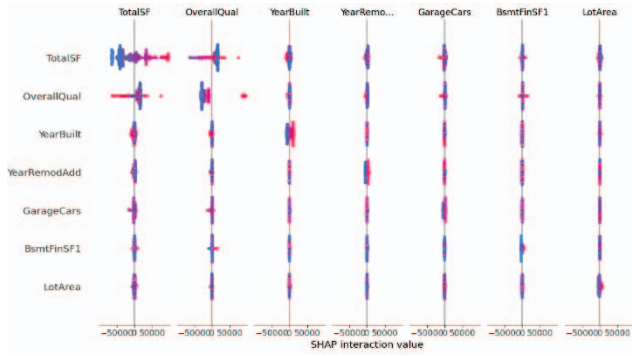


FIG. 3. SHAP variable interaction value for the Stacking Regressor Model for the top 7 variables.

D. Stacking Model Explainability

To enhance model interpretability and provide insights into feature contributions in the Stacking Regressor model, this study employs Shapley Additive exPlanations (SHAP), a widely recognized technique for model explainability (FIG. 2). SHAP values are derived from cooperative game theory, offering a robust framework for understanding the impact of each feature on individual predictions. By calculating the contribution of each feature to the predicted outcome, SHAP values facilitate a comprehensive understanding of how numerous factors influence model decisions. The SHAP framework decomposes the output of a model by the sums of the impact of each feature. SHAP calculates a value that represents the contribution of each feature to the model outcome. These values can be used to understand the importance of each feature and to explain the result of the model to a human.

This approach was integrated into the analysis to understand effects of various features on price predictions, thereby providing transparency into the model's decision-making process. Additionally, SHAP's capability to visualize feature importance through summary plots and dependence plots allowed for detailed exploration of feature interactions and their contributions to the prediction model. This method not only enhances the interpretability of complex models but also aids in identifying potential areas for model improvement and ensuring that predictions align with domain knowledge. Through the application of SHAP, the study aims to bridge the gap between model accuracy and interpretability, offering clear and actionable insights into the factors driving house price estimations.

III. RESULTS

The RFR model demonstrated the highest r^2 score of 0.909, indicating that it explains approximately 91% of the variance in house prices. Its performance was further reflected in a MAE of 16,350.11 and an RMSE of 24,294.61 (Table I). The GBR model closely followed with an r^2 score of 0.906. It achieved a MAE of 15,979.86 and an RMSE of 24,736.03 (Table I). These results confirm GBR's robust performance in capturing the underlying patterns of the data, although slightly less accurate than RFR in terms of RMSE. SVR achieved an r^2 score of 0.901, with a MAE of 16,427.00 and an RMSE of 25,410.23 (Table I).

While the SVR model also performed well, its higher MAE and RMSE compared to RFR and GBR indicate it was less effective in minimizing prediction errors.

In contrast, the TensorFlow Keras NN model showed significantly lower performance, with an r^2 score of 0.423, a MAE of 45,545.78, and an RMSE of 61,278.54 (Table I). These metrics suggest that NN struggled with capturing the complexities of the data, leading to larger prediction errors. The Stacked Regressor, which combines predictions from the multiple regression base models, achieved an r^2 score of 0.901, a MAE of 15,550.72, and an RMSE of 27,529.61 (Table I). The lower MAE indicates an improvement in predictive accuracy.

A. Feature Analysis

The XAI aspect of the stacked model reveals what variables were the most important in the prediction process. The SHAP analysis reveals several key insights. The feature 'TotalSF' (total square footage) demonstrates the highest impact on the model's prediction, with a strong SHAP value with the target variable (FIG. 3). The 'OverallQual' feature, which represents the overall material and finish quality of the house, also shows significant positive contributions (FIG. 3).

'YearBuilt' and 'YearRemodAdd' are indicative of the house's age and any remodeling activities (FIG. 3). Both features show effects, with newer houses or recently remodeled houses contributing to higher predicted prices. 'GarageCars,' representing the number of cars that can fit in the garage, is another critical feature, showing a positive interaction with house prices.

'BsmtFinSF1', which indicates the finished square footage of the basement, also contributes positively, highlighting the value added by finished basements. Lastly, 'LotArea' shows a moderate positive impact, indicating that larger lots can enhance property values, although to a lesser extent compared to the previously mentioned features.

The SHAP interaction plots further detail the interaction effects between pairs of features (FIG. 3). For instance, the interaction between 'TotalSF' and 'OverallQual' suggests that houses with larger square footage and higher quality materials significantly boost predicted house prices. Several of the relations between the underlying variables display relationships, including 'TotalSF' and 'YearBuilt,' 'OverallQual' and 'YearBuilt,' and 'BsmtFinSF1' and 'OverallQual.' These relationships can represent underlying relationships the model utilizes to make its final predictions.

TABLE I. MODEL PERFORMANCE METRICS

Model	Result Metric		
	r^2	MAE	RMSE
RFR	0.909	16350.11	24294.61
NN	0.423	45545.78	61278.54
GBR	0.906	15979.86	24736.03
SVR	0.901	16427.00	25410.23
Stacking Regressor	0.901	15550.72	27529.61

IV. DISCUSSION

Comparing the results of the baseline Regression models and the neural network, the regression models performed well, while the neural network had poor results. The RFR model is highly effective in explaining the variance in house prices and minimizing prediction errors. The GBR demonstrates comparable robustness to the RFR model. The slight difference in RMSE between RFR and GBR indicates that while both models capture underlying data patterns effectively, the RFR model provides marginally better error minimization. The SVR model was also observed to perform well, however, the higher MAE and RMSE compared to RFR and GBR indicate that SVR, while robust, is less effective in minimizing prediction errors. The neural network underperformed significantly compared to the regression models. This stark contrast underscores the neural network's limitations in capturing the complexities of the dataset, leading to substantial prediction errors.

On the contrary, the Stacked Regressor model was also strong in its predictive performance. The model proved to be very comparable in its r^2 to the regression models. More interestingly, the model provided a lower MAE, indicating an improvement in predictive accuracy throughout the individual predictions. On average the predicted house prices were closer to the actual house price than the baseline models. These observations highlight the benefits of model stacking.

The analysis of Explainable AI (XAI) within the context of the stacked model provides crucial insights into the factors driving house price predictions, enhancing the interpretability and transparency of the model's decisions. The SHAP analysis reveals key variables that significantly impact the model's predictions, offering a deeper understanding of the underlying relationships between features and house prices.

The SHAP values highlight the paramount importance of 'TotalSF' (total square footage) in predicting house prices. This aligns with domain knowledge, where larger homes inherently possess higher market values. SHAP values quantify the contribution of each feature, making it evident that 'TotalSF' holds the highest impact, thereby confirming its critical role in the model's predictive capability.

'OverallQual,' representing the overall quality of materials and finishes, emerges as another significant feature. The high SHAP values associated with 'OverallQual' emphasize the model's reliance on construction quality to determine house prices. This underscores the relevance of building quality as a substantial predictor in real estate valuation models.

The features 'YearBuilt' and 'YearRemodAdd' further illustrate the model's sensitivity to temporal aspects of property valuation. SHAP values for these features indicate that newer houses or those with recent renovations contribute positively to the predicted prices. This finding reflects market dynamics where the age and condition of a property, indicative of modernization and upkeep, play crucial roles in determining market value.

'GarageCars,' indicating garage capacity, shows positive interaction with house prices. This feature's impact highlights the added value that amenities such as larger garages bring to residential properties.

Finally, the feature 'LotArea,' representing the size of the lot, also shows a positive impact on property values, though to a lesser extent compared to other features. This finding aligns with the understanding that larger lots offer more space and potential for development, thereby enhancing the property's appeal and value in the market.

The SHAP interaction plots provide additional depth by detailing how pairs of features interact to influence predictions. For instance, the interaction between 'TotalSF' and 'OverallQual' suggests that houses with larger square footage and higher quality materials significantly boost predicted house prices. This synergistic effect demonstrates that the combined influence of multiple high-impact features can elevate property valuations.

The stacked model's integration of multiple regression base models offers notable improvements in predictive accuracy, as evidenced by its lower MAE compared to individual models. This enhancement suggests that stacking effectively leverages the strengths of various models, leading to more accurate and reliable predictions. The comprehensive approach of the stacked model ensures that predictions are not only accurate but also interpretable, thanks to the insights provided by SHAP analysis.

V. FUTURE WORKS

Several avenues for future research can improve the implementation of XAI frameworks in stacked modeling for house price prediction. One promising direction is the integration of more advanced neural network architectures, such as deep learning models, into the stacked model's architecture to capture further complex relationships within the data. Additionally, expanding housing datasets to include more real estate markets and varied economic conditions could enhance the model's generalizability and predictive ability. Implementing more refined feature selection and dimensionality reduction methods might also improve model performance and interpretability in the future. Furthermore, exploring hybrid models that combine traditional regression techniques with modern deep learning approaches could yield potential improvements. These models could leverage the strengths of both methodologies, potentially leading to more accurate and robust predictions.

Advancing XAI methodologies to explore the impact of other interpretability tools, such as LIME (Local Interpretable Model-agnostic Explanations) or integrated gradients, could provide deeper insights into model behavior and feature importance in unique variable relations that may not appear in traditional explainability visualizations. Implementing such XAI stacking based prediction models in user-friendly interfaces that are accessible can help individuals with limited access to real estate knowledge to plan and explore in different investor and residential markets.

VI. CONCLUSION

The comparative analysis between regression models for house price prediction, and a Stacking Regressor Explainable AI (XAI) model, demonstrates significant advancements in both predictive accuracy and model interpretability. In contrast, the TensorFlow Keras Neural Network (NN) significantly underperformed, highlighting its limitations in handling the

complexities of the dataset. The incorporation of SHAP analysis within the stacked model framework offered insights into feature importance, emphasizing the critical roles of ‘TotalSF,’ ‘OverallQual,’ ‘YearBuilt,’ and other key variables. These findings underscore the necessity of feature interpretability in predictive modeling, ensuring that stakeholders can trust and understand the model’s predictions.

The successful application of XAI and model stacking in this study highlights their potential to revolutionize predictive analytics in real estate and beyond. By elucidating the contributions of individual features, XAI enhances the transparency and reliability of complex models. The ensemble approach of model stacking leverages the strengths of various models, resulting in more accurate and robust predictions.

In summary, the integration of advanced ensemble techniques and XAI represents a significant step forward in predictive modeling. Future research should focus on refining these methodologies, incorporating more diverse datasets, and exploring hybrid models to further enhance predictive accuracy and model interpretability for real estate price prediction. Ongoing development and application of XAI and ensemble models promises to drive innovation and improve decision-making processes across numerous fields, making these tools valuable for future research and practical applications.

VII. ACKNOWLEDGEMENT

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VIII. APPENDIX

The project code was published on GitHub for open-source access to improve this work in the future. More figures and information can be found in this project code. Please contact the first author for the code.

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